Intelligent Parking Lot Application Using Wireless Sensor Networks

Sangwon Lee, Dukhee Yoon, Amitabha Ghosh

University of Southern California, Los Angeles, CA, USA

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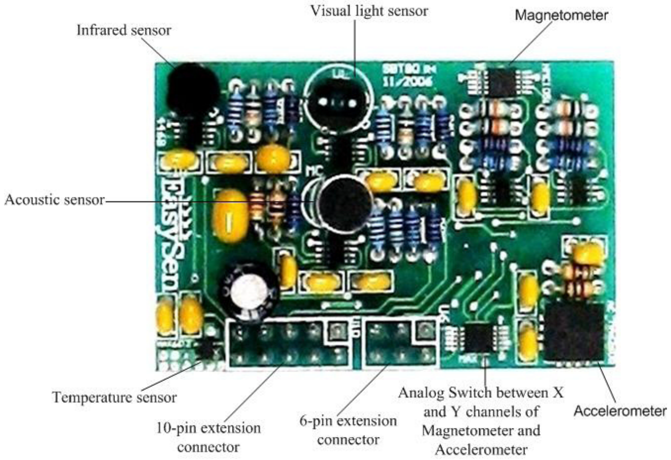
Agenda

- Introduction
- Device Specifications
- Sensing Experiments
- Detection Algorithm
- Transmission Experiment
- Car Counting Experiment
- Conclusion & Future work

Introduction

- Objectives
  - To Count the number of vehicles on each floor
  - Cheap and accurate solutions
- Existing approaches
  - A wide-area sensor network architecture in which video cameras, microphones, and motion detectors.
  - one sensor node for each parking spot
  - A sensor nodes glued on the pavement
- Our approach
  - A combination of magnetic and ultrasonic sensors

Device Specifications (1) Magnetometer



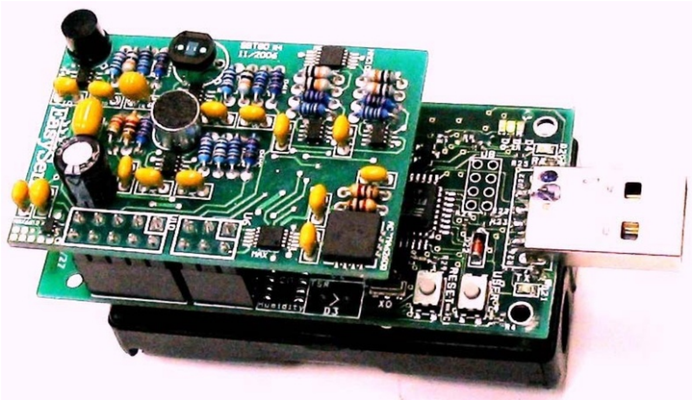
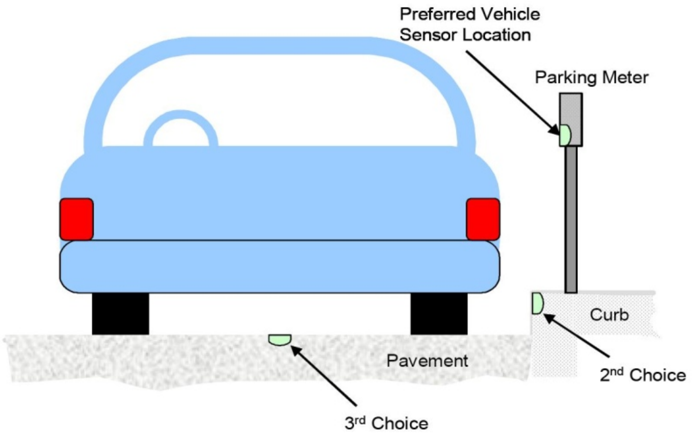


Table 1. Sensors on SBT80 Board

Sensor Type	Manufacturer, Model
Visual Light	PerkinElmer, VTB9412B
Infrared	Advanced Photonix, Inc., PDB-C139F
Acoustic	Horn, EM6050
Temperature	Maxim Integrated Products, MAX6612MXK
Magnetometer	Honeywell, HMC1052
Accelerometer	Freescale Semiconductor, MMA6260Q



Device Specifications (2)
Ultrasonic sensor

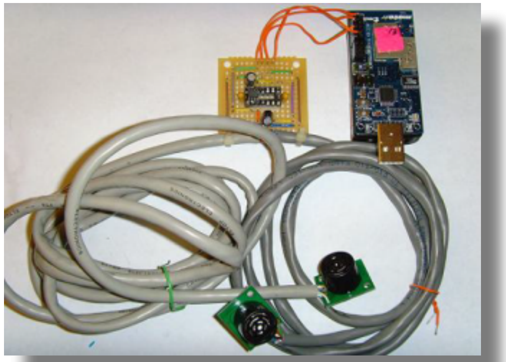


Table 2. SRF02 Ultrasonic Sensor Properties

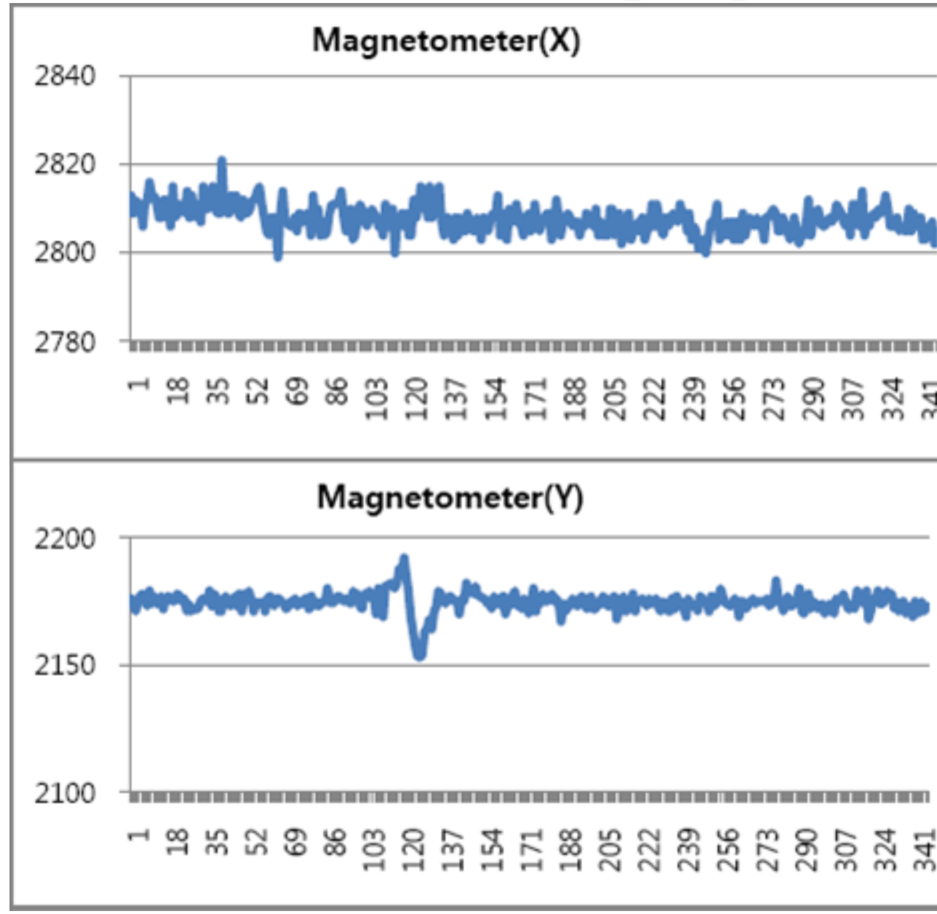
Protocol	I2C, Serial
Range	15cm - 6m
Response Time	70ms
Frequency	40kHz
Voltage	5v
Current	4mA

Sensing Experiments (1)

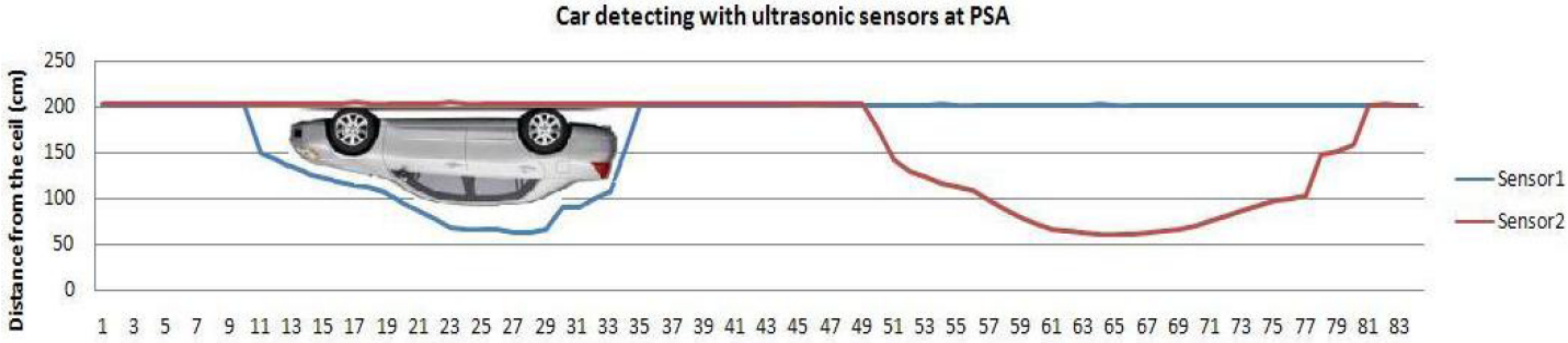
(a)

(b)

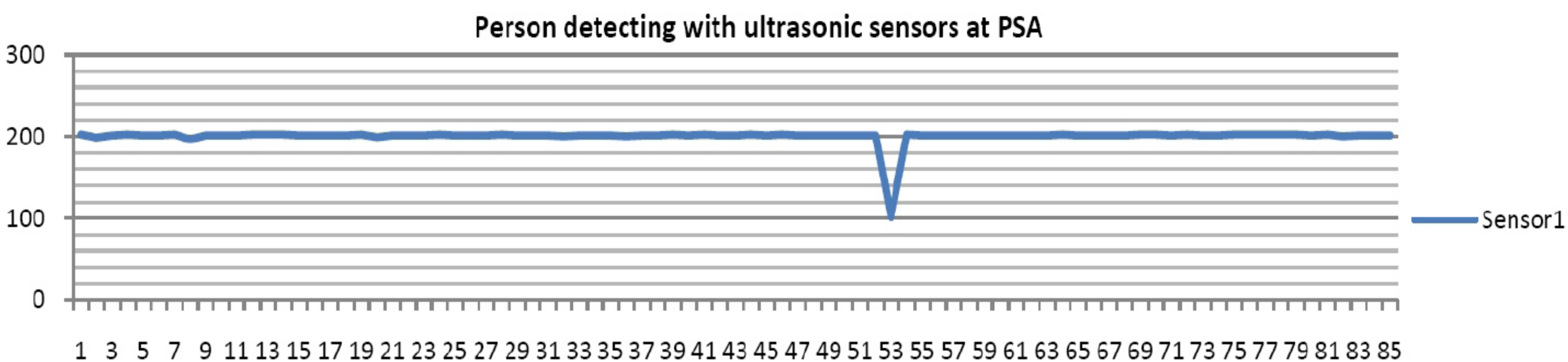
Figure 5. Test Bed: (a) Parking Structure Inside USC Campus. (b) A Tmote Mounted With SBT80 Placed Near the Entrance.



Sensing Experiments (2)
Ultrasonic sensor



(a)



(b)

Detection Algorithm (1)

Modified Min-Max A

Sensing by magnetometer

↓ rawData

Local average

↓ localAvg = 0.9*localAvg + 0.1*ra

Squaring the difference between local average and current raw data

↓ e = (rawData- localAvg)^2

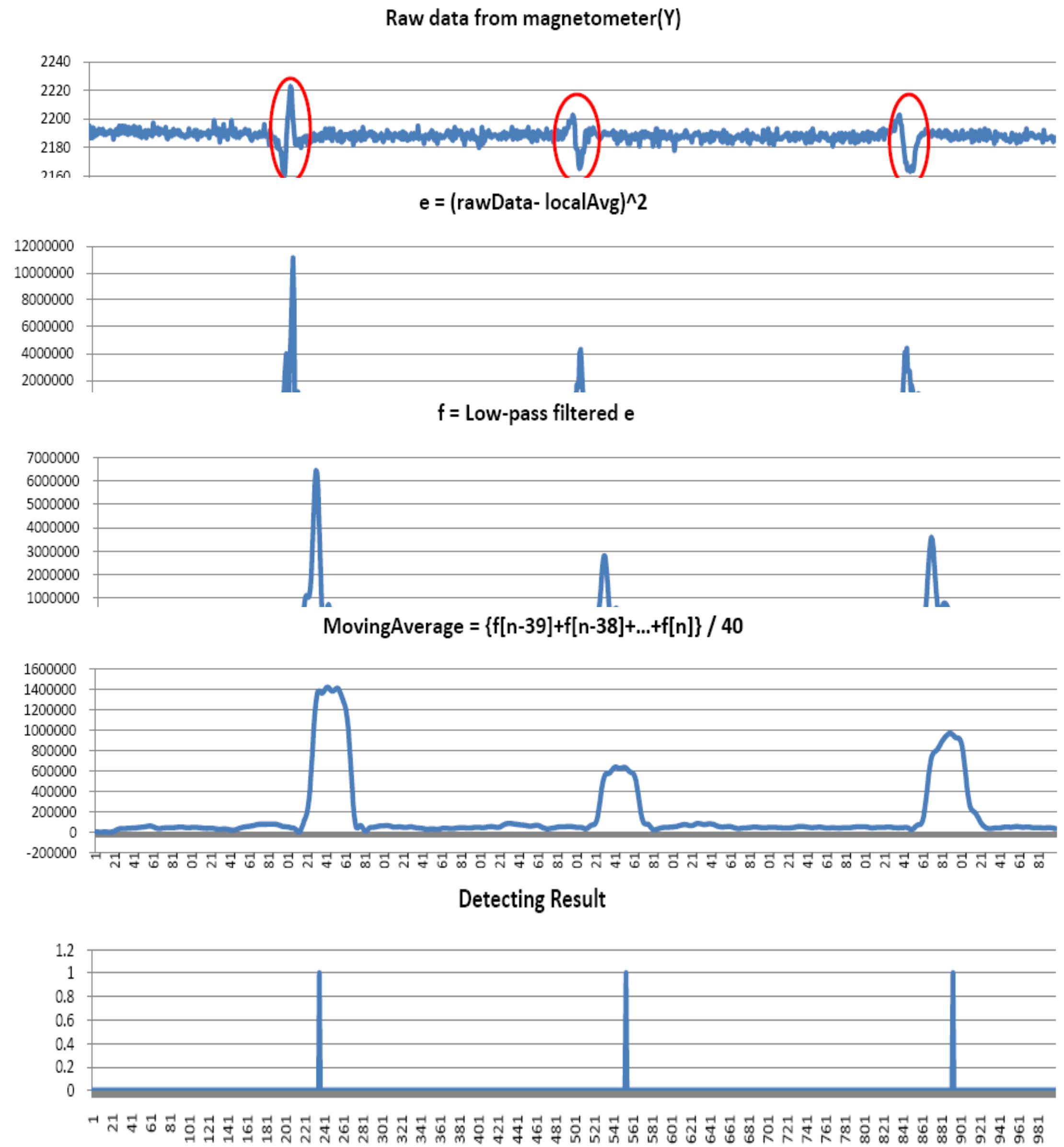
Low-pass filtering

↓ f = Low-pass filtered e

Moving average

↓ MA = {f[n-39]+f[n-38]+…+f[n]} /

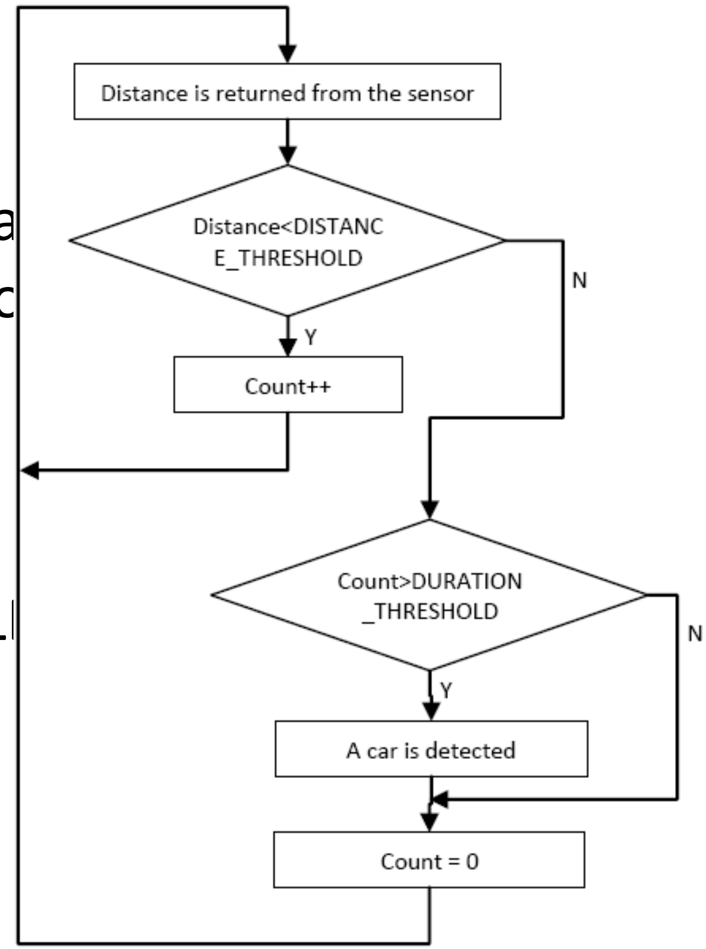
Min-max detection



Detection Algorithm (2)

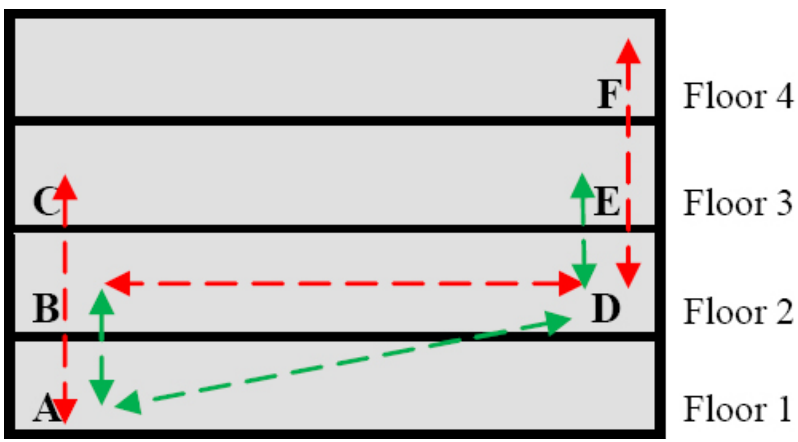
Ultrasonic Sensor Detection Algorithm

- Detecting algorithm
  - DISTANCE_THRESHOLD
    - If the distance returned is sma DISTANCE_THRESHOLD, the cc zero.
  - DURATION_THRESHOLD
    - The algorithm decides a car if Count>DURATION_THRESHOLI



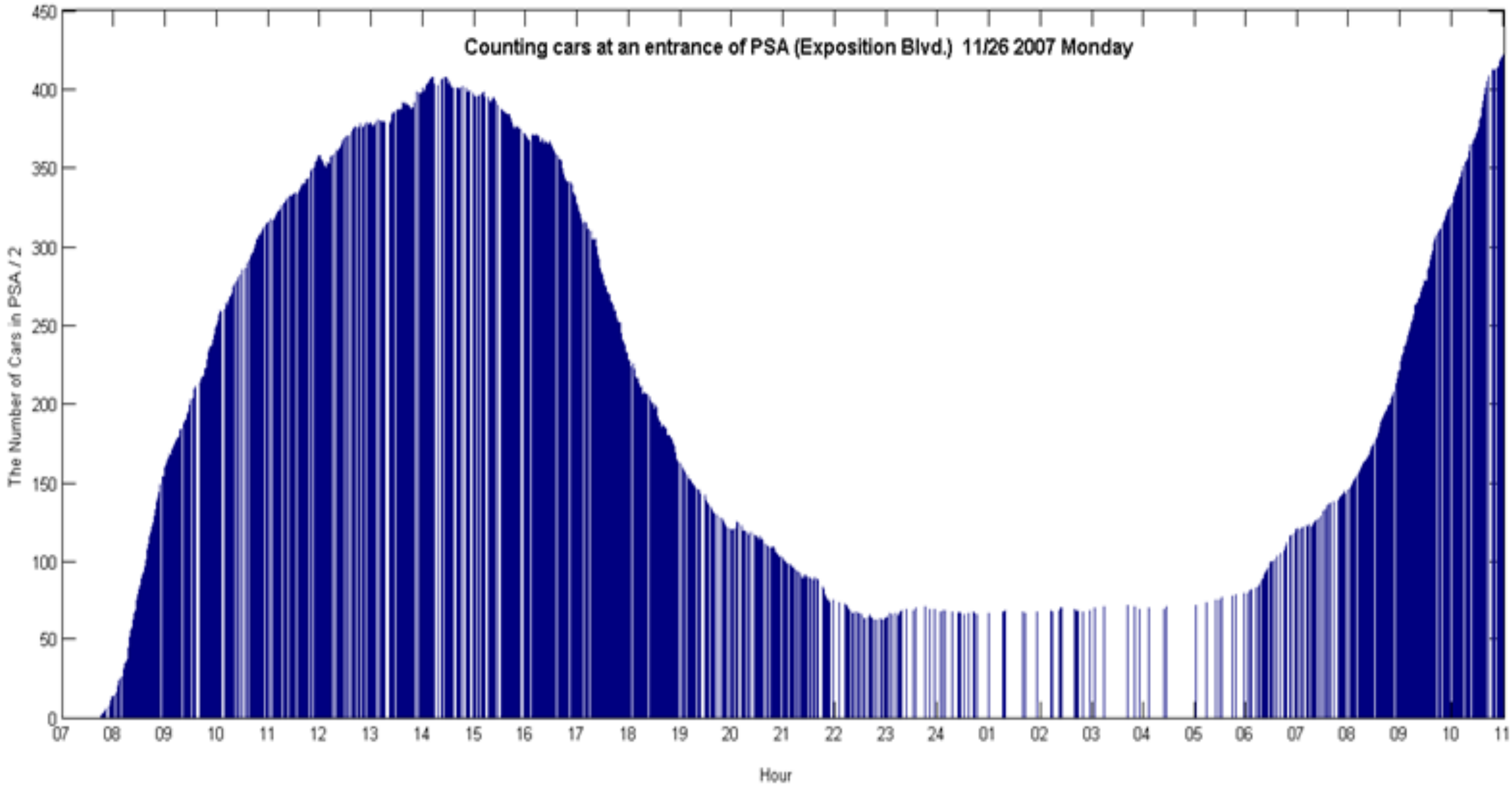
Transmission Experiments

- Transmission experiment
  - Extra routers/forwarders for reliable
  - Delayed transmission



Link	PRR	RSSI
A - D (along the roadway)	0.67	-88.35
A - B	0.72	-85.64
D - E	0.54	-85.01
A - C	0	none
D - F	0	none
B - E	0	none
C - F	0	none

Car Counting Experiments



Conclusions & Future Work

- Most effective of sensors
  - Magnetometer in SBT80 sensor board
  - Ultrasonic sensor, SRF02
- How to be more accurate
- How to be more energy-efficient

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Thank You!